

Monetary policy and the benefits and limits of central bank independence

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Disclaimer: The views expressed in this paper are those of the author and do not necessarily reflect the views of colleagues, the Reserve Bank of New Zealand, and other members of the Reserve Bank of New Zealand Monetary Policy Committee.

Abstract: Central bank independence is an important development in macroeconomics. It has become a lynchpin of modern central banking. The period since the Global Financial Crisis and the experience of the Covid-19 pandemic era posed new challenges for central banks, with potential implications for central bank independence. This paper reviews the scope and rationale for central bank independence and the reasons it spread internationally after the 1980s. It reviews evidence of the effects of central bank independence on inflation, output volatility, financial stability and fiscal policy. The paper considers political threats and contemporary policy issues that could influence the form of central bank independence and credibility. These issues include matters relating to assessments of monetary policy during the pandemic, the scope of central bank mandates, interpretations of inflation dynamics, the funding of central bank operations, and the interaction between fiscal and monetary policy.

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Monetary policy and the benefits and limits of central bank independence

1. Introduction

It is a great pleasure to be invited to speak at the Annual Conference of the New Zealand Association of Economists (NZAE). I attended my first NZAE conference during the mid-1970s. At that time, average annual inflation for OECD countries was persistently around 10 per cent, and rising. Four years later it was over 15 per cent and hovered around 8 per cent until the early 1990s. Thereafter, inflation declined and never exceeded 8 per cent for over 30 years, and persisted for long periods between 1 and 3 per cent, until the resurgence after the Covid-19 pandemic.

The containment of inflation for over 30 years in many countries was not a chance event. A transformation in central banking took place during the 1990s. Many countries gave greater operational independence to central banks for monetary policy, supported by novel accountability and transparency requirements. This occurred initially among developed economies, although many developing economies also soon adopted this institutional framework for monetary policy. During this process, inflation declined globally. While other factors contributed to low inflation, central bank independence (CBI) came to be regarded as the crucial foundation for credible monetary policy and persistent low inflation.

Central bank independence is an important development in macroeconomics, and is a lynchpin of modern central banking. Nevertheless, confidence in CBI has recently come under strain and in some instances been threatened. Some claim central banks have become far less independent than two decades ago.¹ Others are concerned with the expanded role of central banks and implications for their clarity of purpose, adequate accountability and legitimacy.² Some have suggested central banks have become less focussed on inflation.³ Others have expressed concern over rising political threats to central bank independence, or that the constituency for low inflation has been eroded.⁴

In extending this invitation, the NZAE Council suggested I speak on the broad developments of a topic in economics or economic policy. In the spirit of that request, I will discuss the background to the emergence of CBI, review its scope and meaning, its spread internationally, and why it was adopted by so many countries. I also will review evidence of the benefits arising from CBI. Finally, I will address some issues that have emerged in recent times relevant to the concerns mentioned above and which have potential implications for central bank independence and monetary policy.

¹ See for example, Rogoff (2022), Lingle (2021)

² See for example, Tucker (2018a), Rajan, (2023).

³ See *The Economist* (2022).

⁴ See Goodhart & Lastra (2018a) on rising political threats. King (2023) suggests the constituency for low inflation is eroding.

2. The scope and growth of central bank independence

Central bank independence refers to the freedom of monetary policymakers from direct political or governmental influence in the conduct of policy.⁵ This does not involve complete independence from government because legislation specifying the objectives of a central bank are determined by governments. Furthermore, funding, board appointments, and accountability requirements of central banks are determined by governments.

There are different elements to CBI and the manner in which they are specified varies across central banks. These elements cover how the central bank aligns with the political process and other organs of government, the financing of a central bank, and the nature of the employment conditions of central bank governors and monetary policy committee members. They cover how the specification of central bank objectives are determined and how policy decisions are formed and approved. They also cover issues relating to the choice of policy instruments (degree of instrument autonomy), and accountability requirements.⁶

To provide structure to this complexity, attempts have been made to measure the degree of independence using the laws of a country that establish and govern them.⁷ The most comprehensive attempt are the coding and weighting rules developed by Cukierman, Webb and Neyapti (1992) (CWN), shown in Table 1. The CWN scheme became a widely accepted basis for *de jure* measures of CBI (although others have been developed and used for research).

Table 1: The CWN coding and weighting rules to measure central bank independence

Dimension of independence	Components (Weight in index)	Variables (Weight in the component)	Categories
Personnel	CEO/Governor (0.20)	Term of office of CEO (0.25)	5
		Who appoints the CEO (0.25)	5
		Provisions for dismissal of CEO (0.25)	7
		Rules about CEO holding another office in government (0.25)	2
Objectives	Objectives (0.15)	Central Bank objectives (1)	6
Policy	Policy formulation (0.15)	Who formulates monetary policy (0.25)	4
		Government directives and resolution of conflicts (0.50)	6
		Central Bank role in government budget formulation (0.25)	2
Financial	Limitation on lending to government (0.5)	Limitations on advances (0.30)	4
		Limitations on securitized lending (0.20)	4
		Who decides control of lending terms to government ((0.20)	4
		Beneficiaries of Central Bank lending (0.10)	4
		Type of limits when they exist (0.05)	4
		Maturity of loans (0.05)	4
		Restrictions on interest rates (0.05)	5
		Prohibition on Central Bank lending in primary market to government (0.05)	2

Notes: Adapted from Garriga & Rodriguez (2020) and is a summary of Table 1 in Cukierman, *et al.* (1992).

⁵ Walsh (2009).

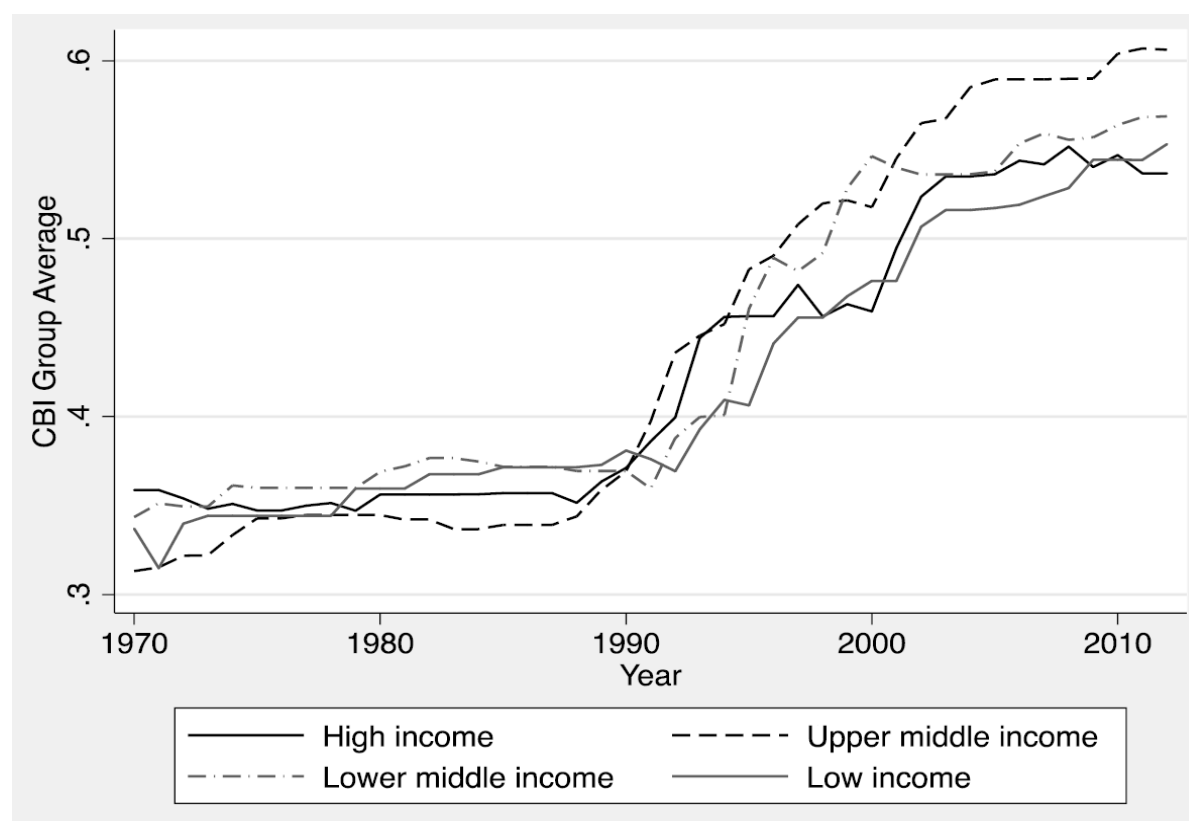
⁶ Bernanke (2017).

⁷ These *de jure* measures (i.e., based on legislation) of central bank independence trace back to Bade & Parkin (1978). Their approach was to specify four categories of CBI. Subsequent approaches derived indices of CBI (Parkin, 2013). Examples are Alesina (1988), Grilli *et al.* (1991), Alesina & Summers (1993), Fry, *et al.* (2000). The development of these indices is reviewed in Eijffinger & de Haan (1996) and Arnone *et al.* (2006).

The most comprehensive application of the CWN rules for measuring CBI appears to be that by Garriga (2016) covering 182 countries over the period 1970 to 2013. Measures of CBI derived by Garriga for different country-income groups are shown in Figure 1. A striking feature is a surge in the degree of central bank independence after 1990, for all income groups.

Garriga's research also reveals the sources of change in central bank independence. There are substantial differences in the contributions of the four components in Table 1. The largest contributions to change in CBI have been from central bank objectives (especially a stronger focus on inflation), operational independence, and financial independence (this refers to limitations on lending to governments; it does not refer to budgetary independence). Personnel independence has also increased but not by as much as the other three components.

Figure 1: Central bank independence: annual averages for country-income groups



Source: Garriga (2016, Figure 5.9). Online data appendix available from publisher's website <https://www.tandfonline.com/journals/gini20>

Measures of CBI based on statutes have some weaknesses. Statutes do not cover all contingencies that could affect the relationships between governments and central banks, and they can miss relevant influences on central bank independence. An example is the absence in the CWN coding (and other *de jure* measures) of a measure of restrictions on funding of some central bank monetary policy operations (discussed in Section 5.6). The weightings attached to each component of central bank laws can be contested. Nevertheless, *de jure* measures have the advantage of providing common measures of independence over time and across countries. Attempts have been made to derive *de facto* measures of independence (e.g., the turnover rate of central bank governors, and measures based on surveys of central bankers). But *de facto* measures also have significant weaknesses, including the difficulty of establishing a consistent measure across countries and over time.⁸

3. Reasons why central bank independence spread internationally

What prompted this dramatic change in the institutional framework for monetary policy during the 1990s? After all, it seems the performances of early independent central banks were mixed, and the concept had sceptics among influential monetary economists. According to Goodfriend (2012), independent monetary policy originated in the 19th century with the Bank of England and the US Federal Reserve. He considers the Bank of England's lender of last resort policy served its purpose well, but the Federal Reserve ran monetary policy according to an ill-defined public interest and became a source of macroeconomic instability. In contrast, the independent Bundesbank and Swiss National Bank were relatively successful at achieving persistently low inflation during the post-WW2 decades. This was influential, especially in European Monetary System countries.⁹

An early contribution to the CBI debate was by Milton Friedman (1962) in his discussion of monetary structures "both stable and free from irresponsible government tinkering". He considered three alternatives: a commodity standard, an independent central bank, and legislated rules for monetary aggregates. Friedman was sceptical that an independent central bank could deliver low inflation consistently. He argued it was rare for an independent central bank to not take account of government preferences, and the degree of independence was dependent on the personality of central bank governors. He preferred instead that government legislate rules for the instruments of policy, and specifically for money supply growth.

A dramatic change in Federal Reserve policy took place under the leadership of Paul Volcker. In October 1979 the focus shifted toward controlling the money supply with the aim of reducing inflation.¹⁰ Annual inflation had risen from 3 to 9 per cent in 18 months and was projected to

⁸ Cukierman *et al.* (1992) first proposed turnover rates of governors. Although often used in subsequent studies, the suitability of turnover rates has been questioned by Dreher *et al.* (2008, 2010) who found the turnover rate is subject to endogeneity issues. Furthermore, the turnover rate of governors does not take account of the degree of independence of other members of a monetary policy committee. An example of measures of CBI based on surveys of central bankers is Fry *et al.* (2000). This was based on a one-off survey responded to by central bank staff. Such surveys are not easily replicable and rely on the accuracy of responses which could be influenced by what staff believe or would prefer, or would like others to believe.

⁹ See Cukierman (1994). Alesina & Summers (1993) suggest the historical experience of hyperinflation in Germany may have raised German public aversion to inflation and its interest in a more independent central bank committed to price stability.

¹⁰ The Federal Reserve Reform Act of 1977 had earlier required the Federal Reserve to direct its policies toward achieving maximum employment and price stability and report progress to Congress (Federal Reserve System, 2022).

rise further. By the end of 1983, the annual inflation rate had been brought down to 4 per cent, and low inflation was sustained until 2021.

The Volcker episode showed that monetary policy could achieve a significant reduction in inflation. However, the output and employment costs of this policy were accentuated by weak policy credibility.¹¹ The system was still vulnerable to Friedman's concern of dependence on the personality of central bank governors. The problem was similar to the consequences of the time-inconsistency of optimal plans highlighted by the seminal work of Kydland and Prescott (1977). To overcome this problem and the inflationary bias it created in the context of monetary policy, subsequent literature emphasised the importance of expectations, the credibility of monetary policy, and the benefits of enforced commitments (or rules) over discretion.¹² To achieve a credible commitment to low inflation, Rogoff (1985) proposed delegating monetary policy to an independent central bank in order to insulate central bankers from political pressures that could otherwise induce elected politicians to renege on promises of low inflation.

It is widely considered that it was not until New Zealand introduced in 1989 a formal arrangement combining operational independence and inflation targeting supported by a high degree of transparency and accountability, that the idea of CBI (and inflation targeting) took flight. This innovation was introduced in New Zealand as part of a wider set of public sector reforms and a search for a solution to persistent high inflation.¹³ A feature of the legislation was a contractual relationship between the central bank Governor and the Government that changed the incentives of the central bank in a manner consistent with Rogoff's proposal.¹⁴ Many features of New Zealand's approach influenced central bank governance in other countries. By the late 1990s, a consensus was emerging that the primary aim of monetary policy should be to control inflation and CBI was an effective way to do this.¹⁵

Other factors influenced the spread of central bank independence. Research revealing the adverse effects of persistent high inflation motivated the search for a solution to breaking high inflation.¹⁶ Public-sector debt ratios in many countries had become much lower in this period which meant that when central banks raised interest rates to reduce inflation, it would have been of less concern for governments.¹⁷ The potential for CBI to signal creditworthiness to foreign investors in an environment of more internationally integrated financial markets may have also been a motivation.¹⁸ The influence of particular interest groups such as the financial sector which was expanding during the 1990s, may therefore have been important. The expansion of democratic regimes during the 1980s and 1990s may have been important. It has

¹¹ Goodfriend & King (2005).

¹² For example, Barro & Gordon (1983) and Backus & Driffill (1985) and earlier work by Milton Friedman, Robert Lucas and others in emphasising the critical role of expectations.

¹³ Evans, *et al.* (1996), Grimes (1996). The CBI reforms were enacted by the Reserve Bank of New Zealand (RBNZ) Act 1989.

¹⁴ The influence of the NZ legislation is mentioned by numerous commentators. Examples are Mishkin & Savastano (2001), Goodhart & Lastra (2018b), Loungani (2023). Walsh (1995) concluded the principal-agent approach implicit in the novel accountability aspects of New Zealand's approach was designed to sustain a socially optimal commitment policy.

¹⁵ Bernanke & Mishkin (1997).

¹⁶ High inflation in the 1970s and 1980s prompted more research into the effects of inflation on economic growth and how those effects are manifest. See for example Grimes (1991) and Johnson (1993) and the extensive literature cited in these papers.

¹⁷ Goodhart & Lastra (2018a).

¹⁸ Maxfield (1997).

been suggested democracies are more predisposed to the decentralisation of policy decisions, and therefore more likely to allow central banks to pursue price stability objectives without political interference.¹⁹ The IMF was active promoting the virtues of CBI in its Article IV country reviews. A relatively benign global macroeconomic environment after the 1980s is also likely to have supported the spread of CBI and inflation targeting.

4. What have been the benefits of central bank independence?

The growth of independent central banks after 1990 was associated with a shift in focus of monetary policy toward inflation targeting.²⁰ Initial interest therefore focussed on the marginal effects of CBI on inflation and, to a lesser extent, output volatility. The effectiveness of monetary policy also depends on the efficiency and stability of financial systems and the cooperation of fiscal authorities. Accordingly, CBI spurred interest in evaluating whether politicians circumvent the discipline of monetary policy independence via financial system regulations and fiscal policy. The research into the marginal effect of central bank independence has evolved considerably and benefitted from the availability of larger samples of countries, improved data on central bank independence, improved econometric estimation techniques and understanding of the importance of other factors that can influence the impact of central bank independence. The following is a flavour of the insights from this research.

4.1. Central bank independence and inflation

Evaluation of the marginal impact of CBI on inflation goes back to Bade and Parkin (1978), followed by several other studies in the 1990s.²¹ The broad conclusion was that CBI resulted in lower inflation, with variation by country coverage and choice of indices of independence. These studies focussed on developed economies and samples were relatively small. Subsequent research extended country coverage, tested components of *de jure* independence and *de facto* measures of independence, searched for possible coefficient estimation bias due to not adequately controlling for other influences on inflation, allowing for country specific effects, and improved econometric methods including dynamic panel estimation methods.²² This large volume of research was evaluated by Klomp and de Haan (2010a) using meta-analysis. Although they found results varied by time period, the size of inflation rates, and country coverage, their overall conclusion is that CBI reduces inflation.

Significant effects of CBI reducing inflation were found in studies for Latin American countries, democratic developing countries, former communist countries, and in countries with initial high rates of inflation. Nevertheless, a common finding from the studies reviewed by Klomp and de Haan is that while the effect of CBI on inflation was found in small subsets of

¹⁹ Lim (2021)

²⁰ Bernanke *et al.* (2001, p4) defined inflation targeting as “a framework for monetary policy characterised by the public announcement of official quantitative targets (or target ranges) for the inflation rate over one-or-more time horizons, and by explicit acknowledgement that low, stable inflation is monetary policy’s primary long-run goal.”

²¹ These include studies by Alesina (1988), Grilli, *et al.* (1991), Cukierman, *et al.* (1992), Alesina & Summers (1993), and in a similar way by Carlstrom & Fuerst (2009).

²² Some of the variables included in these studies to account for other influences on inflation include the exchange rate regime, degree of international openness, fiscal policy, size of inflation rates, and characteristics of labour market conditions.

developing countries, the effect in developing countries was not always evident.²³ Garriga and Rodriguez (2020) suggest these differing results may have been due to sample selection bias and estimation methods. They revisit this question using a sample of 118 non-OECD countries covering 1980 to 2013, using the CWN approach to measure the degree of CBI.²⁴ They find that in most developing countries, a higher degree of CBI results in lower inflation. The effect is stronger in the presence of more democratic institutions, in the presence of a floating exchange rate, and when a central bank operates under an explicit inflation targeting mandate. The effect of CBI on price stability nevertheless persists after controlling for these effects.

The spread of CBI facilitated greater use of monetary policy instruments to supplant the prevalent use of less effective nonmonetary devices to stem the persistent high inflation that characterised the decades prior to the 1990s.²⁵ This is likely to have been an important factor enabling CBI to achieve sustained lower inflation.

4.2 Central bank independence and output volatility

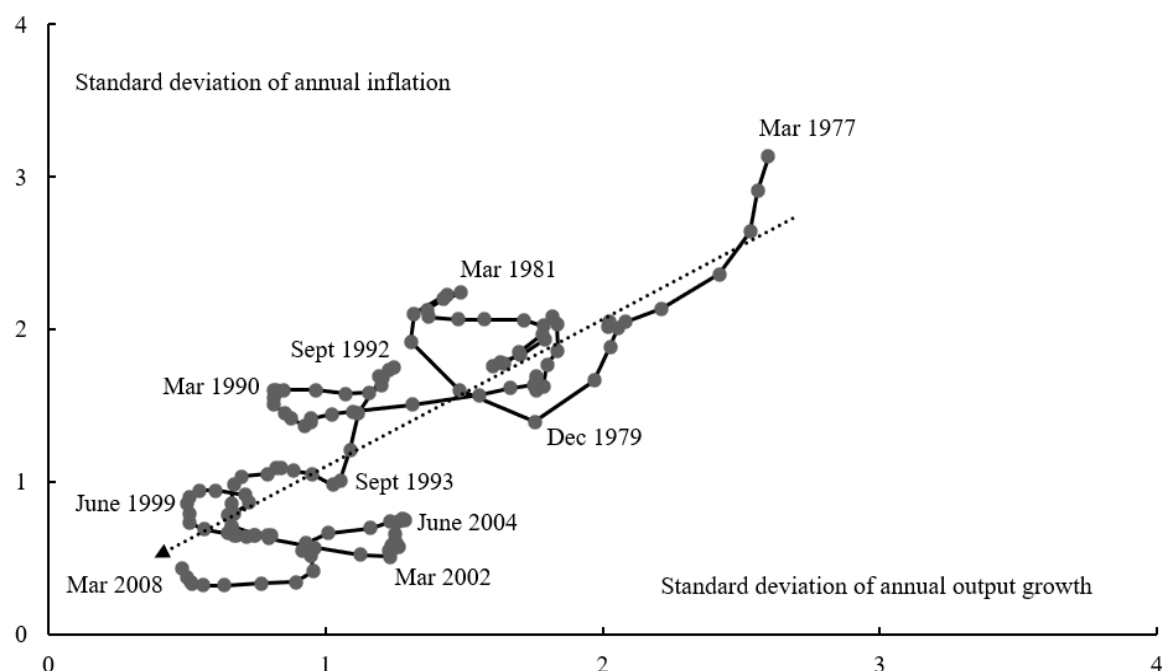
The decades following the initial surge of CBI were characterised by falling volatility of inflation and output growth, coined the “Great Moderation” by Bernanke (2004). Figure 2 plots at quarterly intervals, the combinations of standard deviations of average annual inflation rates and real output growth for OECD countries. It reveals that volatility of both inflation and output growth were generally higher during the late 1970s and the 1980s. During the subsequent two decades, when central bank independence was spreading globally, inflation and output volatility were on a declining trend. There are cycles about this trend. In the early 2000s, output growth volatility increased (possibly due to spill-over effects of the Asian Financial Crisis and the ‘Dotcom’ share market crash). Thereafter, inflation and output volatility continued to decline, at least until the Global Financial Crisis.

²³ These various studies can be found in Jácome & Vázquez (2008), Bodea & Hicks (2015), Loungani & Sheets (1997), Acemoglu, *et al.* (2008), Landström (2011), and Klomp & de Haan (2010b)

²⁴ Although their conclusions are robust to alternative *de facto* measures of CBI.

²⁵ See discussions and research in De Long (1997), Romer & Romer (2002), Bernanke (2004), and Nelson (2005, 2022).

**Figure 2: The “Great Moderation”:
OECD inflation and output growth volatility from 1977 to 2008**



Notes: Author's calculations using OECD data. Each data point is the final year of five-year moving average standard deviations for the average of OECD countries' annual inflation and real GDP growth. Dotted line is an estimated linear trend.

Several factors contributed to this development. These include China's sustained high growth of low-cost manufacturing output exported to the rest of the world, demographic factors resulting in a rising proportion of the working age population, rising labour force participation rates, and technological improvements to supply-chain management and inventory control.²⁶ Changes in monetary policy associated with CBI have also been credited as contributing to the decline in macroeconomic volatility (Bernanke, 2004; Taylor, 2013). By strengthening credibility and changing inflation expectations, central bank independence and inflation targeting change the dynamics of inflation. This alters the trade-off between inflation and output volatility, and therefore macroeconomic volatility.²⁷

4.3 Central bank independence and financial stability

The intersection between monetary and financial conditions means CBI could impact financial stability, though there is debate as to whether CBI fosters or weakens financial stability. Arnone, *et al.* (2009) and Bernanke (2010) have suggested CBI fosters financial stability. On the other hand, Atkin and Kern (2020) suggest increased CBI resulted in weakening of financial regulation in some countries, as governments sought to compensate for the loss of control of monetary policy and find other ways to expand credit for politically strategic areas.

²⁶ A review of research on contributions to declining output volatility during the “Great Moderation” is in Buckle, *et al.* (2003).

²⁷ This point is articulated by Bernanke (2004) who couched his argument within the context of Taylor's (1979) monetary policy efficiency frontier. The way improved central bank credibility can change inflation dynamics and improve the Taylor efficiency frontier is explained in Clarida, *et al.* (1999), Orphanides & Williams (2005a), and Buckle (2019) who also reviews evidence of the impact of CBI and inflation targeting on New Zealand's inflation and output volatility.

There are various ways CBI and inflation targeting could influence financial stability. A central bank that credibly committed to low inflation may generate more predictable interest rates and lower long-term interest rates and therefore interest rates and asset prices will be more stable.²⁸ There are counter-arguments. An independent central bank that has established a credible low inflation policy may reduce the probability investors and financial institutions assign to economic downturns, which might encourage higher risk investment behaviour.²⁹ The combination of CBI and inflation targeting raised concerns this could result in more abrupt interest rate changes that pose higher financial risks for financial institutions.³⁰ Recent USA bank failures is illustrative of what can happen when policy interest rates are increased rapidly.

The empirical research on the effect of CBI on financial stability is more recent, but there is an emerging theme which is that central independence can improve financial stability. Several studies indicate CBI is associated with improved banking stability.³¹ Others evaluate the impact of CBI on financial stability more broadly using time-series panel models that control for other potential influences. For instance, Klomp and de Haan (2009) cover 75 developed and developing countries from 1985 to 2005 and find that CBI reduces financial instability. Moreover, countries in which central banks have an explicit mandate for financial supervision have improved financial stability. Fouejieu (2017) evaluates developing countries using a composite macro-financial condition index covering the period 2001 to 2010. The results suggest inflation targeting central banks are relatively more sensitive to financial risks.

These studies covered years prior to the Global Financial Crisis (GFC). The GFC challenged the view that inflation targeting central banks will give sufficient consideration to financial stability. There are a few studies that take account of the GFC period. These also suggest that CBI contributes to lower financial instability. They show that the effect of CBI in moderating financial instability is more important in countries with high levels of democracy.³²

4.4 Central bank independence and fiscal policy

Granting independence to a central bank involves politicians foregoing economic policy flexibility. The question that naturally arose when CBI was spreading was whether it would induce more fiscal discipline by governments, or whether the objectives of monetary policy would be compromised by fiscal dominance. If CBI was considered by politicians as part of a broader package of reforms to achieve sustained low inflation and fiscal discipline is deemed part of that broader package, greater fiscal discipline might be an outcome. If on the other hand politicians consider low inflation the sole preserve of the independent central bank thereby easing a constraint on fiscal policy, there may be less fiscal discipline. Alternatively, political parties that have inherited CBI legislation but consider repeal is not politically prudent, may

²⁸ See Baltensperger (1993), Cukierman (1994), Schwartz (1995).

²⁹ See Borio & Lowe (2002).

³⁰ See Cukierman (1992) and Mishkin (1996), while Berger & Kirsmer (2013) explore the potential implications for financial stability of the different incentives of politically determined and politically independent monetary policy decisions.

³¹ For example, García-Herrero & Lopez (2003) and Čihák (2010).

³² See for example Doumpos, *et al.* (2015), Andries, *et al.* (2022) and Nguyen & Dang (2022).

try to use fiscal policy to circumvent the aims of CBI. These motives could result in “fiscal dominance” reducing the effectiveness of CBI.³³

Fiscal dominance can occur when there is conflict over stabilisation policy that results in a competitive game that frustrates the central bank’s ability to pursue its monetary policy goals.³⁴ It can also occur is when governments generate unsustainable fiscal deficits and public debt that influence expectations of the likelihood of future monetisation of public debt and higher inflation. Perceptions that fiscal policy and public debt levels are not sustainable can therefore compromise current and future monetary policy.³⁵

An early investigation into the relationship between CBI and government budget deficits by Parkin (1987) suggested countries with independent central banks had smaller deficits, although this was not conclusive. Subsequent studies using different indices of central bank independence and different samples of developed and developing countries spanning the years from 1950 to 1992 (years prior to the surge of CBI and low inflation) produce a mix of significant and insignificant effects of CBI on fiscal deficits.³⁶ However, these studies may not adequately capture the full effects of CBI which became more widespread after the early 1990s.

There are several studies that cover the years since the expansion of CBI. In general, these studies are more robust in controlling for other influences on fiscal outcomes and suggest budget deficits tend to be smaller when central banks are more independent, but there are important qualifications that have emerged from this research.³⁷ These studies suggest the impact of CBI on fiscal policy is conditioned by domestic political and legal institutions. The influence of CBI on fiscal policy is more effective in democracies and in countries with robust legal systems, a free press, and constraints on the political executive, and less effective in autocratic regimes. The effect appears stronger for those countries that are part of a currency union and share a common central bank, than for countries that have a national central bank. A higher level of financial development also improves fiscal performance and reduces reliance on central bank financing. These studies find evidence of a political cycle effect within democracies. CBI is more likely to influence fiscal balances outside election years.

4.5 Summary of effects of central bank independence on macroeconomic performance

Broad conclusions of the marginal effects of central bank independence are provided in Table 2. Judgement has had to be used to generate these conclusions. They are drawn from studies using different research methods, different country samples, a mix of stages of economic development, different political regimes, and using different measures of CBI and financial instability, and there is a mixture of conclusions from these studies.

³³ The basis for considering monetary dominant and fiscal dominant regimes is developed by Leeper (1991) and discussed in the context of recent events by Leeper (2023).

³⁴ For example, Buckle & Stemp (1991).

³⁵ For example, Sargent & Wallace (1984).

³⁶ These papers and results are summarised by Strong & Yayi (2021, Table 1, p4).

³⁷ These conclusions and qualifications are drawn from studies by Bodea & Higashijima (2017, which includes post-1990 years for 78 countries that include democracies, more autocratic regimes and dictatorships), Burdekin & Laney (2016, which covers Central and South American countries from 1990 to 2012), Strong & Yayi (2021, for 30 African countries from 1990 to 2017).

The strongest conclusion is that CBI reduces inflation, while the weight of evidence suggests CBI also reduces financial instability and fiscal deficits. There are qualifications to these conclusions. The effects of CBI on inflation, financial stability and fiscal deficits tend to be stronger in democratic regimes, and the effect on fiscal deficits is enhanced in countries with independent legal and media institutions (and some evidence a currency union matters). The evidence points to a stronger effect of CBI where the objectives are specified in legal mandates and the central bank is subject to strong accountability requirements. These studies do not cover the Covid-19 pandemic era, when inflation and macroeconomic volatility increased markedly.

Table 2: The marginal effects of central bank independence

Macroeconomic condition	The marginal effect of CBI	Selection of qualifications			
		Type of economy	Other institutional factors	Political cycle effect	Do explicit mandates matter?
Inflation	Reduces.	Developed and developing.	Stronger effects in: Democracies, flexible exchange rates, more internationally open. Labour market institutions matter.	na	Yes. Inflation targeting also practiced in some countries without new legislation.
Output volatility	Reduces. Conclusion from theoretical models and evidence of CBI effect on inflation expectations. Limited direct empirical research.	Developed. Limited evidence available for developing economies.	na	na	Unclear. But inflation targeting regimes tend to reduce output volatility.
Financial instability	Reduces. Conclusion for periods before and after the GFC. There are contrary results and contested views.	Developed and developing	More effective in democracies	na	Yes. Those with IT or FS mandates more sensitive to financial risks.
Fiscal deficits	Reduces. There are qualifications. More conclusive in samples that include the post-1980s.	More effective with independent legal and media institutions. Financial development important.	More effective in democracies.	Yes. The effect of CBI on fiscal deficits is weaker in election years.	Fiscal frameworks are influential. Stronger effect for countries in a currency union.

5. Some contemporary issues for central bank independence

In spite of these benefits, there are issues that could threaten or impact the characteristics of central bank independence. These include political threats to CBI, the scope of central bank mandates, funding arrangements, dissatisfaction with central bank responses to the Covid-19 pandemic, and the relationship between monetary and fiscal policies.

5.1 *Threats from overt political pressures*

The status of independent central banks has come under threat from political movements that are discontent with the status quo. Goodhart and Lastra (2018a, p50) suggest this threat is particularly acute when democratically elected politicians try to remove the “checks and balances” generally applied in a democracy. During the 2017 and 2022 French Presidential campaigns, the candidate for the National Rally Party (Marine Le Pen) was advocating changes to the policy decision making processes and institutions that were interpreted as a threat to the independence of the Banque de France.³⁸ Criticism over timing of response to the post-pandemic inflation prompted an unusually explicit threat by Pierre Poilievre, leader of the Conservative Party of Canada and of the Parliamentary Official Opposition, to remove the Governor of the Central Bank of Canada if elected.³⁹ During the 2016 USA Presidential campaign the Republican candidate (Donald Trump) publicly criticised the then Federal Reserve Chair. As President, he launched a series of public criticisms of the decisions of the Federal Reserve as he embarked on an expansionary fiscal programme.⁴⁰

The independence of central banks in these countries has to date survived these threats, but this is not the case in Turkey. The coalition Government formed after the 2002 elections and led by Prime Minister Erdoğan (and later President) started to lose support after the GFC. As the economic situation deteriorated and democratic processes were eroded, and as the costs of losing power increased, pressure came on the Central Bank of Turkey by the President and his Government to lower interest rates.⁴¹ Having repeatedly stated his belief that high interest rates cause inflation, the President dismissed two central bank governors. Under the new Governor Kavcıoğlu, successive interest rate cuts were implemented. Annual inflation soared to 85.5 per cent by October 2022, and the value of the Turkish lira plummeted.⁴²

The events in Turkey illustrate how central bank monetary policy decisions can be vulnerable to political pressure. The ultimate outcome in Turkey is extreme. Nevertheless, even in

³⁸ Goodhart & Lastra (2018a, p58).

³⁹ Rutter (2022).

⁴⁰ In 2018 he remarked that the Federal Reserve is “my biggest threat...Because the Fed is raising rates too fast. And it’s independent...” Cited from “Trump says Fed is his ‘biggest threat’ because it is raising rates too fast” (cnbc.com).

⁴¹ Demiralp & Demiralp (2019) discuss and analyse the erosion of democratic processes and systematic pressure placed on the Central Bank of Turkey. They find that commentaries from the President and Cabinet Ministers advocating a lower policy interest rate had a significant influence on the Bank of Turkey’s policy reaction function.

⁴² Based on official statistics. By April 2023 the rate had reduced to 43.68 per cent. In the early 2000s, under CBI, annual inflation in Turkey fell from between 60 and 105 per cent to below 10 per cent. Evidently two factors are helping prevent a more severe currency crisis following the interest rate reductions and soaring inflation: an influx of funds from Russian depositors unable to deposit in other countries, and the government is seeking large transfers from Arab nations (Hardie, 2022).

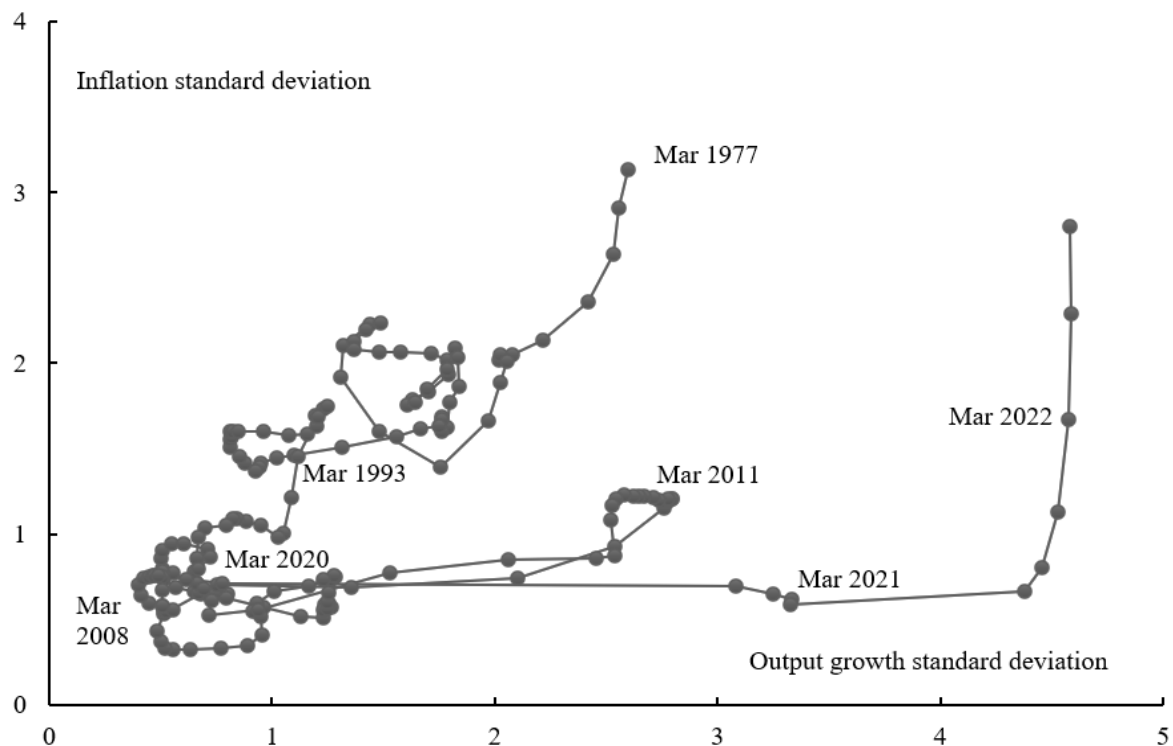
countries where there is overt political support for central bank independence, there can be insidious developments that have implications for the nature of central bank independence.

5.2 Deteriorating macroeconomic performance and central bank reputations

Favourable macroeconomic conditions may have helped build a constituency of support for CBI and inflation targeting in the early decades, but those conditions have been less favourable since the GFC. While the GFC temporarily increased macroeconomic volatility in many countries, inflation was contained. However, macroeconomic volatility increased even more abruptly during the pandemic era. This is illustrated by Figure 3 which extends Figure 2. Output volatility surged in 2020, and a surge of inflation volatility soon followed. Macroeconomic volatility reached levels not previously experienced in the era of central bank independence.

The Covid-19 pandemic posed unprecedented challenges for governments and central banks. In a review of the Australian Government's policy responses, Borland (2023, p66) suggests the defining characteristic of the pandemic is "huge and ongoing uncertainty": There was uncertainty over the epidemiological features of the virus, its impact on peoples' health, how society would respond to government health mandates, the frequency and duration of those mandates, how the financial system would react, and the effectiveness of unique fiscal measures introduced to buttress economies. Initial forecasts for economies were dire.⁴³

Figure 3: Macroeconomic volatility from the "Great Moderation" to Pandemic, OECD



Notes: Author's calculations using OECD data. Each data point is the final year of five-year moving average standard deviation for the average of OECD countries' annual inflation and real GDP growth.

⁴³ For example, the Secretary to the Australian Treasury remarked in 2020 that in forecasting the effects of the pandemic, falls in Australian GDP of 20 per cent were being "seriously contemplated." (Kennedy, 2020). Another example of the degree of uncertainty was the reaction of equity markets. The percentage falls in the S&P500 index on 12 and 16 March 2020 were similar to the falls on 28 October and 8 November 1929 (Hawkesby, 2020, p7).

Central banks were faced with unique circumstances and enacted unprecedented policy responses. There are variations in the particular features of monetary policy instruments used but overall, they were aimed to lower interest rates and keep them lower, stabilise financial markets and support the flow of credit.⁴⁴ Fiscal measures to support employment and household incomes were effective and labour markets proved more adaptable than expected, helping to support household incomes.⁴⁵ Asset prices surged, further supporting aggregate demand. The uncertainties of the pandemic were compounded by Russia's invasion of Ukraine in February 2022 further disrupting supply-chains causing higher energy and food prices globally. As a result of an unexpectedly strong rebound of aggregate demand, enduring disruptions to supply-chains and commodity price rises, inflation rose quickly to rates not experienced for 30 years in many countries.

Central banks have been widely criticised for their responses to these events.⁴⁶ *Ex post* assessments of performance during unusual periods like this are important for accountability and insights to guide future policy.⁴⁷ There have been some attempts to evaluate central bank decisions, but they have been partial only. It is not sufficient to assess monetary policy decisions solely on the basis of how policy interest rates compare with estimates of “neutral” policy rates as some critics have done. Nor is it sufficient to simply identify demand-induced and supply-induced contributions to inflation to assess the impact of monetary policy and fiscal policy.⁴⁸ A robust assessment requires taking account of central bank mandates and the trade-offs required to sustain price stability over the short and medium terms, the information available at the time decisions were made, including knowledge of the scope and effects of government fiscal and health responses, as well as the efficacy of the policy responses.

The perception of the performance of central banks during this period will have a bearing on the tolerance for central bank independence. If central banks are to retain public confidence in their independent status, it is important they respond to criticism and promote research that identifies behavioural responses to monetary policy during the pandemic. This work should evaluate the design and scope of policy initiatives including the size of policy rate cuts, the duration and specification of forward guidance, the design and scope of quantitative easing and other measures introduced to support the funding and liquidity of financial institutions, and the timing of policy reversals. Insights from this research can help central banks adapt their

⁴⁴ The unique economic circumstances and considerable uncertainties faced by central banks during the outbreak of Covid-19 and the policy responses and thinking behind those responses are discussed in various central bank speeches and papers such as for example Orr (2020) and Debelle (2020b) for small open economies, and for the USA see Clarida, *et al.* (2021).

⁴⁵ Although the comparison of US and UK policies by Tenreyro (2021) illustrates there were significant difference among countries in the characteristics of the fiscal measures and the resultant employment outcomes.

⁴⁶ For example, Rogoff (2022) has criticised the timing of the US Federal Reserve Board's increase in interest rates and curtailing of quantitative easing. Wheeler & Wilkinson (2022) claim that central banks became complacent about their ability to maintain low inflation and are critical of the delays reversing accommodative monetary policies. Borio (2023) suggests central banks initially underestimated the strength and persistence of inflationary pressures.

⁴⁷ An example of the benefits of *ex post* reviews is the evaluation of Australian fiscal policy during the Covid-19 pandemic is available in Borland (2023).

⁴⁸ Shapiro (2020 and 2022) for example, identifies demand and supply shocks to inflation by classifying demand-induced components of inflation as those where an unexpected change in price moves in the same direction as the unexpected change in quantity in a given month; supply-induced components are identified as those where unexpected changes in price and quantity move in opposite directions. The New Zealand Treasury (2023) and Reserve Bank of Australia (2023) have replicated this approach using New Zealand and Australian data respectively. Some critics have used this type of decomposition to judge the performance of fiscal and monetary policies during the pandemic era.

decision processes to be better prepared for future events that threaten inflation and macroeconomic stability. This is particularly so for complex events like the pandemic, that impact the supply-side of economies (production and labour markets.)⁴⁹ Some central banks have started this process and have acknowledged mistakes in timing, magnitude and duration of their policy responses during the pandemic.⁵⁰

5.3 Implications of state-dependency of inflation dynamics for policy strategy

One of the criticisms of central banks during the pandemic is they were slow to react to emerging signs of inflation after the pandemic and many expected inflation impulses from supply-chain disruptions would be temporary (or transient). One reason suggested for this is that central banks' views of inflation dynamics were conditioned by the post-GFC years.⁵¹ This earlier period was dominated by persistent low inflation, frequent occurrence of inflation falling below target bands and fear of disinflation.⁵² Instead of acting to pre-empt inflation, some central banks signalled during the early stages of the pandemic they would wait until inflation had clearly materialized before taking action.⁵³

Research in the 1980s and 1990s produced insights suggesting inflation dynamics may be 'state-dependent'. For example, research revealed asymmetric pricing behaviour that was systematically related to inflation. One reason for this is that if it is costly to change price, firms will delay changes until the private benefits outweigh the private costs. As inflation rises, the net benefit to firms of raising prices increases.⁵⁴ The consequence is that firms raise prices more frequently as inflation rises, thereby changing the dynamics of inflation.

The more recent idea of 'rational inattention' proposed by Sims (2003) is another reason inflation dynamics could be state-dependent. The idea of is that people cannot process all available information, but they can choose which exact pieces of information to attend to. In the context of price-setting and the formation of inflation expectations, it could be rational to pay more attention to inflation when it is high than when it is low. This implies inflation expectations and therefore wage and price setting behaviour could be more responsive to

⁴⁹ This type of challenge is not new. Central banks struggled to understand the reasons for persistent low inflation and undershooting of inflation targets during the decade following the GFC.

⁵⁰ Examples are Reserve Bank of New Zealand, 2022; Kashkari, 2023. The recent review of the Reserve Bank of Australia (Australian Government, 2023) assessed the monetary policy framework, decision-making processes, use of macroeconomic models, and forecasting methods and has made sweeping recommendations.

⁵¹ For example, Brunnermeier (2023).

⁵² The persistence of very low inflation and the challenges this posed for inflation targeting central banks is discussed in Tarullo (2017), Constancio (2017), and Spencer (2017). This period prompted suggestions that inflation was a concern of the past and inflation targeting and CBI was no longer relevant, a suggestion that has been refuted by several, including Draghi (2018).

⁵³ Most notably the Federal Reserve Board (FRB) and the Reserve Bank of Australia (RBA). The FRB approved a new policy framework on 27 August 2020 with the Statement on Longer-Run Goals and Monetary Policy Strategy. This included five elements including one stating "The Committee expects to delay liftoff from the ELB until PCE inflation has risen to 2 percent on an annual basis and other complimentary conditions, consistent with achieving this goal on a sustained basis and to be discussed later, are met." The statement includes other elements suggesting monetary policy would remain accommodative for some time (Clarida, 2022). The RBA initially provided state-dependent forward guidance in March 2020, stating that "The Board will not increase the cash rate target until progress is being made towards full employment and it is confident that inflation will be sustainably within the 2–3 per cent target band." The RBA added a time-dependent component in October 2020 and November 2020, and reiterated this in February 2021 by indicating a time horizon of 3 years over which the cash rate would be unlikely to change (Australian Government, 2023, pp 41-42).

⁵⁴ Some of the theoretical and empirical literature is contained in Sheshinski & Weiss (1993). See also Ball & Mankiw (1994). An example of empirical evidence that the frequency of price rises increases with inflation is in Buckle & Carlson (2000).

inflation as it rises.⁵⁵ An implication for monetary policy is that while it may be desirable in some circumstances to tolerate moderate deviations from inflation targets, pre-emptive monetary policy becomes more important as the risk of transition to high-inflation increases.⁵⁶

A further body of emerging research examining the pandemic era of inflation in the USA suggests it may be important to pay particular attention to the initial source of inflation in order to understand the dynamics of inflation. Inflation pressures can arise from product or labour markets, and the ultimate nature of the inflation dynamics can depend on where the original source of inflation pressure arose, the duration of that pressure, how product and labour markets interact, and how expectations are formed.⁵⁷

The fact inflation dynamics, including expectations, can be state-dependent is likely to be relevant to the way monetary policy forward guidance instruments should be applied and the selection of the appropriate monetary policy strategy. When unique events like the Covid-19 pandemic occur, using long-duration time-dependent forward guidance could be problematic. This type of policy strategy could ultimately undermine central bank credibility if the dynamics of inflation prove to be different to what the central bank assumed. This is another issue central banks should evaluate when assessing their responses to the pandemic.

5.4 The scope of central bank monetary policy mandates

The expanding scope of central bank responsibilities in recent years has raised concern they have evolved to the point where there are too many objectives and conflicts of interest. Central bank monetary policy mandates are appropriately determined by governments, but as more objectives are included in mandates and more conditions are attached to those objectives, more trade-offs are involved.⁵⁸ This can be particularly problematic if a central bank is expected to achieve a specific inflation objective but also contribute to less well-defined objectives. The recent review of the Reserve Bank of Australia highlighted risks created by broad central bank mandates. The review of monetary policy from 2016 to 2019 concluded that the vagueness of including “economic prosperity and welfare” in its mandate meant the Board could draw on many reasons outside the domain of monetary policy, for not lowering interest rates. This caused conflict with the Bank’s inflation and employment goals and the review recommended reducing the scope of the mandate.⁵⁹

Some possible implications of broad central bank mandates have been raised by Tucker (2018a, 2018b). His concern is that decisions over trade-offs are based on moral judgements and are inherently political decisions. As mandates expand, a central bank will be drawn into a wider net of inherently political decisions that could threaten central bank independence. Without

⁵⁵ Reserve Bank of New Zealand (2023a) shows the sensitivity of inflation expectations to current inflation has been increasing.

⁵⁶ Borio, *et al.* (2023).

⁵⁷ For example, Bernanke & Blanchard (2023).

⁵⁸ This concern has also been raised in the context of the growing tendency for central banks to undertake a financial regulatory and supervisory role. Lastra & Wood (1999) expressed concern that regulatory supervision independence would give bank supervisors to some extent “the coercive power of the state against private citizens.” Quintyn & Taylor (2002) consider this is a power that has no equivalent with independent monetary policy.

⁵⁹ Australian Government (2023).

clear objectives that can be measured and monitored in a transparent way, Tucker suggests an independent agency's legitimacy would be precarious.

A critical issue is the specification of the primary objectives of monetary policy. These vary among central banks.⁶⁰ For many central banks the sole or dominant primary objective is inflation. Some central banks explicitly specify employment stabilisation as a secondary objective conditional on not compromising their primary objective of price stability.⁶¹ A few central banks have dual primary objectives: an employment related objective coupled with price stability.⁶²

Efforts to resolve the optimal specification of monetary policy objectives using stylised models suggest that it may be welfare enhancing to have at least some weighting on resource utilisation objectives (employment, unemployment rate, or the output gap) in addition to inflation. However, conclusions are mixed: some papers conclude that any weight on employment contributes very little to welfare; some find that under certain conditions it is optimal to put some weight on a resource utilisation objective.⁶³ This research highlights the importance of the type of shocks and preferences for stabilising inflation. In response to demand shocks for example, some models show that monetary policy can stabilise the welfare-relevant resource utilisation gap by focussing on the inflation objective, with only mild improvements to welfare if the central bank additionally targets resource utilisation.⁶⁴ It can also be important to differentiate between the types of supply shocks.⁶⁵

Other considerations are important. One is the way expectations are formed. Theoretical model evaluations of optimal specifications often assume agents form model consistent expectations and implicitly assume agents believe inflation will converge to the target rate. If this condition does not hold, and expectations are formed adaptively, policies that put weight on optimising a measure of resource utilisation run a risk that higher inflation will persist longer and undermine a central bank's credibility to achieve low inflation.⁶⁶ This in turn affects the trade-off between inflation and resource utilisation objectives.

Another issue is optimal policy assignment. Even if monetary policy can mitigate to some degree the adverse effects of supply shocks on resource utilisation, it may not be the optimal policy assignment. Regulatory policies or targeted fiscal measures, may be more efficient. The

⁶⁰ See Wadsworth (2017), Archer (2009).

⁶¹ For example, Bank of England, Bank of Canada, Sveriges Riksbank, Norges Bank, and the European Central Bank.

⁶² Notably the US Federal Reserve and Reserve Bank of Australia, whose legislation pre-dates formal inflation targeting. The Reserve Bank of New Zealand originally had a sole primary objective of price stability. This changed in December 2018 when legislation introduced in addition to price stability, the objective of supporting maximum sustainable employment.

⁶³ For a more extensive discussion of these issues see Rotemberg & Woodford (1997), Clarida, et al. (1999), Blanchard & Galí, (2007), Reserve Bank of New Zealand (2023b), Jacob & Özbilgin (2021).

⁶⁴ See for instance, Clarida, *et al.* (1999), Blanchard & Galí, (2007), Jacob & Özbilgin (2021). Typically, these papers consider the implications for representative households and firms and not potential differences between different types of firms and households. In principle, a demand shock generating this "divine coincidence" property may not be "divine" for all sectors. But monetary policy instruments are typically not designed to differentiate between sectors and hence it is often more efficient to deal with adverse effects of demand shocks for some sectors using other (fiscal or labour) policies.

⁶⁵ For example, Jacob & Özbilgin (2021) find that in the presence of significant cost-push shocks to price and wage inflation, welfare is enhanced when the central bank focusses solely on stabilising inflation.

⁶⁶ Orphanides & Williams (2005a) evaluate related issues.

use of various forms of fiscal measures deployed by governments to mitigate the impact of the Covid-19 pandemic on employment and household incomes is an illustration.

A further issue is measurement accuracy. Insights from organisational theory suggest the determination of weights assigned to objectives should be based on measurement accuracy: more accurately measured objectives should be weighted more heavily.⁶⁷ This is consistent with research showing that the chances of policy mistakes due to poorly measured objectives are reduced if greater weight is placed on more accurately measured objectives.⁶⁸ The rate of inflation is generally able to be measured more accurately than employment or output at the time monetary policy decisions are made. This implies that in a mandate that includes inflation and resource utilisation objectives, the inflation objective should be weighted more heavily, or a lexicographic ordering would be appropriate, requiring the central bank to achieve its inflation objective first, and pursue its other objective(s) subject to achieving this.⁶⁹

Adding objectives to central bank mandates that are not best dealt with by monetary policy risks undermining the credibility of a central bank and diluting the case for independence. These considerations together with measurement issues, imply central bank mandates should put greater weight on price stability than on resource utilisation objectives. Although optimal monetary policy objectives may be shock dependent, central banks cannot shift from one monetary framework to another without risking credibility. This implies it would be appropriate to choose a framework or ordering of objectives for all regimes.⁷⁰ The importance of maintaining low inflation and central bank credibility in order to maximise the contribution of monetary policy to well-being, together with uncertainty over measurement of the maximum level of employment, are reasons the Canadian Government retained low and stable inflation as the primary objective of the Bank of Canada.⁷¹

5.5 Monetary policy mandates and ‘secondary objectives’

It is common to find mandates where primary objectives are subject to secondary objectives. These are intended to impose conditions on how a central bank applies monetary policy instruments, or to provide a signal to the public of the intentions of monetary policy.

For example, consider a secondary condition that requires a central bank to adopt a medium-term perspective toward the achievement of inflation targets. This provides legitimacy for a central bank to adopt a flexible approach to inflation targeting and to “look through” events expected to have a temporary impact on inflation. Similarly, conditions pertaining to the volatility of interest rates and exchange rates could be viewed as encouraging a medium-term approach to targeting objectives. On the other hand, restrictions on variables such as interest rates and exchange rates that are important in the transmission of monetary policy, have the

⁶⁷ Coleman & Dasgupta (2023). Also relevant to the weighting scheme is the extent to which the behaviour of the objectives is correlated. The degree of correlation will depend on the nature of the shocks.

⁶⁸ Orphanides & Williams (2005b).

⁶⁹ A lexicographic ordering is applied by the Bank of England. Similarly, price stability is the primary objective of monetary policy for the Bank of Canada, Norges Bank and the European Central Bank, and consideration of employment is conditioned by the ability to satisfy the primary objective.

⁷⁰ Rajan (2023).

⁷¹ Government of Canada (2021).

potential to limit the discretion over the scope of the use of policy instruments in pursuit of the primary objectives. They may also pose a risk that the public expects the central bank to have responsibility for the behaviour of economic variables it is not equipped to influence with any accuracy or in a sustained way. It is also common for mandates to require central banks to consider the implications of monetary policy for financial stability. Although this makes explicit what monetary policy committees should consider during their policy deliberations, there could be a risk the public interprets this as requiring monetary policy committees to have much broader responsibilities.⁷²

There are some secondary conditions that pose more serious risks for central bank credibility. For example, some central banks (e.g., Bank of England and the European Central Bank) have explicit secondary objectives incorporating an element of environmental responsibility. These are not objectives that monetary policy instruments are designed to influence effectively. They may risk raising public expectations about the role of monetary policy and dilute the case for central bank independence. Central bank monetary policy committees should be tasked with focussing on what is critical to achieving their primary objective(s) and managing the trade-offs appropriate to the primary objectives and for which it has suitable policy instruments.

5.6 Operational independence and alternative monetary policy instruments

Following a long period of declining interest rates, the use of policy interest rates during the Covid-19 pandemic was perceived to be constrained by their effective lower bound. There are several policy instruments available when policy rates are close to their effective lower bound.⁷³ Moderately negative interest rate policy (NIRP) was widely considered, particularly in light of the experience using NIRP in a few countries.⁷⁴ There were however, concerns over institutional (and sometimes legal) constraints and uncertainty over how financial systems would cope with negative rates. Central banks therefore chose to use quantitative intervention measures, supported by forward guidance, to reduce interest rates, influence yield curves, and provide liquidity to the financial sector, to try to achieve their mandates.⁷⁵ This increased the size of their balance sheets, as illustrated by Figure 4.

The measures taken during the pandemic built on and went beyond those employed in the GFC, including non-conventional liquidity support to financial firms, as well as large-scale purchases

⁷² This concern has been raised by Goodhart & Lastra (2018b) in the context of the growing tendency for central banks to undertake financial regulatory and supervisory roles in addition to the traditional monetary policy role.

⁷³ Alternative policy instruments are discussed by Lilley & Rogoff (2019) and include raising inflation targets, counter-cyclical fiscal policy, “helicopter money”, negative interest rate policy (NIRP), and forward guidance on interest rates.

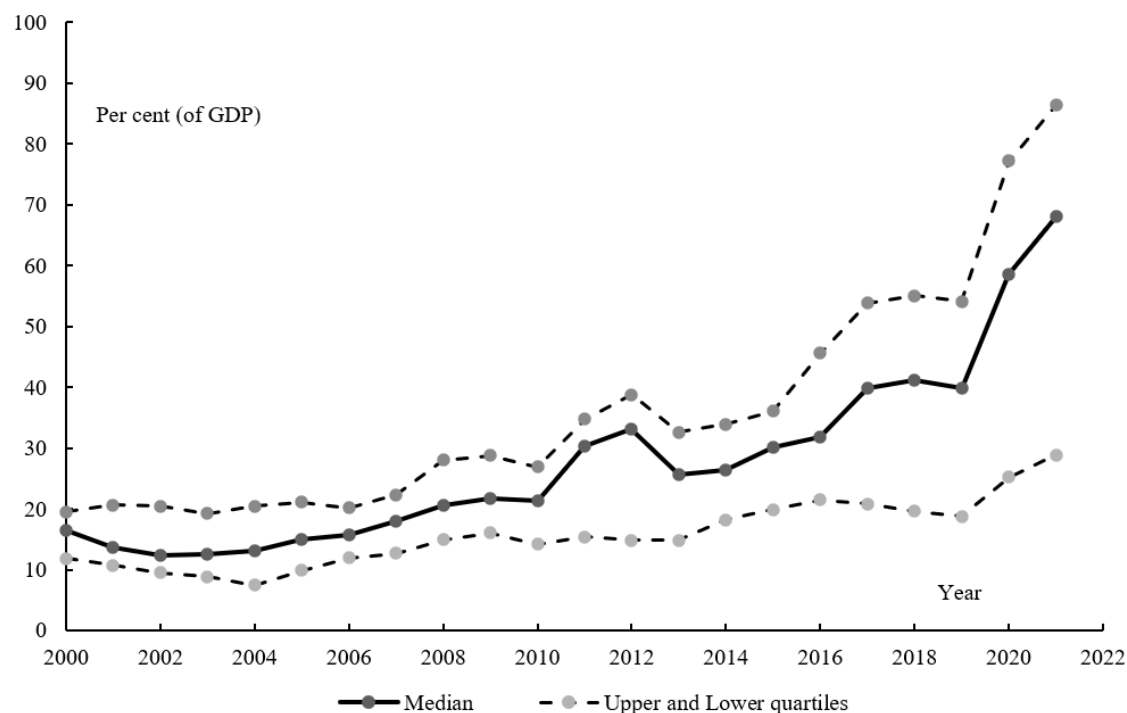
⁷⁴ Negative interest rates were used from 2012 in Europe, Denmark, Sweden, Switzerland and Japan. Reviews by The World Bank (Arteta, *et al.*, 2016) and IMF (Brandão-Marques, *et al.*, 2021) conclude NIRP in these countries supported demand and influenced exchange rates in ways similar to conventional interest rate cuts. Prior concern about the consequences for bank profitability, excessive risk taking, and financial instability did not eventuate materially, although they acknowledge these effects could become more evident if rates were substantially negative or used for longer periods. These reviews stressed the effects of NIRP and extent they can be deepened are likely to depend on the structure of financial systems. Banks and banking systems more reliant on retail-oriented deposit funding for instance, could be more vulnerable to negative interest rates.

⁷⁵ Covid-19 initially triggered a rise in demand for liquidity. In addition to their monetary policy decisions, central banks with financial stability responsibilities bought bonds on secondary markets to improve liquidity and mitigate risks to financial stability. For example, see Hawkesby (2020).

of government securities under quantitative easing.⁷⁶ The expansion of central bank assets was larger in advanced economies where quantitative easing programs caused balance sheet expansion relative to emerging market and low-income countries, where financial sector support was more important.⁷⁷ Central banks typically funded these purchases of government bonds by increasing interest-bearing liabilities, in particular deposits (reserves) held by financial institutions and often remunerated based on the policy interest rate.⁷⁸ While this process will increase base-money, the principal influence is via a change in interest rates.⁷⁹

Bond purchases on secondary markets amounts to transferring risk from private investors and financial institutions onto central bank balance sheets, and therefore also the consolidated central government balance sheet. This process can therefore increase the risk to central bank equity and therefore to operational independence and credibility.⁸⁰ The risks to the central bank will depend on political tolerance for equity losses, market perception of their impact, on a central bank's capital structure, and on loss compensation arrangements with Government.

Figure 4: Central Bank balance sheets, 2000 to 2021:
Total assets as percentage of GDP for 27 advanced economies



Notes: Data for Figure 3 was provided by the authors of Bell, *et al.* (2023) and is a recreation of Graph 1A, p1 in their paper.

⁷⁶ Hooley, *et al.* (2023, p7), who also report that only a few central banks provided direct support to the non-financial sector, and for those that did, the size of support was relatively small in most cases.

⁷⁷ Bullock (2022) discusses reasons for choosing quantitative easing during the pandemic, including a summary of the macroeconomic effects and issues relevant to assessing the impact on the Australian Government's fiscal position. See also Reserve Bank of New Zealand (2022).

⁷⁸ DeBelle (2020a) and Hawkesby (2020) discuss in detail how the RBNZ and RBA balance sheets changed during the pandemic, and how deposits of financial institutions at the central banks are remunerated.

⁷⁹ See for example Silk (2022), Broadbent (2023).

⁸⁰ The financial risks considered here are those that could potentially influence the choice of monetary policy instrument. They do not relate to a central bank's operational budget. The fiscal risks referred to are limited to the direct effects on the central bank and therefore consolidated government balance sheet. There can be other "quasi-fiscal" risks arising from arising from the effect of quantitative easing on the market for public debt and the efficiency government bond programmes (see Hooley, *et al.* 2023)

The financial risk to central bank equity arises from the possibility bond purchases may ultimately result in a central bank incurring net interest income losses. A central bank's holdings of bonds involve fixed interest earnings. If, at a later date, market and policy interest rates rise, interest paid on the stock liabilities on its balance sheet created to finance quantitative easing will also rise, resulting in reduced net interest earnings or losses for the central bank.

The market value of central bank assets also declines with rising bond yields. This also reduces profits of central banks that sell bonds at the lower market value, or that continue to hold bonds and recognise their decline in the market value when calculating net profits. Some central banks (Bank of England, Bank of Canada, RBA and RBNZ) value their bond holdings based on market values and recognise the decline in value as bond yields rise.⁸¹ The impact of net interest rate and market valuation losses on central bank equity will depend on the size and duration of bond holdings and size of the interest rate differential. They also depend on income recognition, rules governing distribution of central bank net income and interest rate risk transfer agreements with the fiscal authority. These conventions and rules vary among central banks.⁸²

While the primary purpose of monetary policy is the efficient pursuit of the central bank's mandate rather than maximising profits from bond trading, risks associated with quantitative easing and bond holdings can have fiscal consequences.⁸³ Therefore, large or sustained operating losses and negative equity can be politically sensitive and may influence public confidence in central bank operations. There may also be financial market confidence effects and perceptions that income and equity losses affect the scope of central bank monetary policy decisions and tolerance for risk (e.g., associated with quantitative easing).⁸⁴

When there is uncertainty about how central bank losses will be dealt with, misperceptions can arise and erode central bank credibility.⁸⁵ Accordingly, some central banks are compensated for losses by indemnity arrangements with the fiscal authority.⁸⁶ Some central banks view indemnities as irrelevant from the perspective of the overall public sector balance sheet and consider they could even reduce policy effectiveness if dependence on indemnities negotiated with government weakens actual or even perceptions of CBI. Some central banks therefore hold financial buffers or additional capital to finance potential net interest losses.⁸⁷ Some rely

⁸¹ Many central banks (including the Federal Reserve, European Central Bank, and Bank of Japan) value them at the amortized value and do not recognise unrealised profits or losses because they anticipate holding them to maturity (Hooley, *et al.*, 2023).

⁸² Bell, *et al.* (2023, p4) note that in advanced economies, historically the net income of central banks has been a modest source of government revenue. But this can vary over time. When policy rates were continuously low following the GFC from 2010 to 2019, central bank income and corresponding transfers to government were relatively large. However, during the pandemic period, net interest losses for several central banks have led to smaller transfers to the fiscal authority or none at all, and in some cases probably for years to come (See Chart 1D, p1).

⁸³ Unlike commercial businesses, central banks are not at risk of insolvency because they are typically protected from court-ordered bankruptcy, and are underwritten by government. Moreover, central banks have the ability to create "base money" to finance operations. There are limits to the scope to generate their own funding despite a monopoly over base money creation, because there is a limit to the demand for liquidity, and because the extent of the recourse to seigniorage required to safeguard a central bank's solvency might be of a scale that could undermine monetary policy objectives (Buiter, 2008; Reis, 2013).

⁸⁴ These and other risks are reviewed and discussed by Nordström & Vredin (2022).

⁸⁵ Bell, *et al.* (2023).

⁸⁶ For example, the Bank of England is fully indemnified by UK Treasury via its subsidiary the Asset Purchase Facility. The RBNZ and Bank of Canada have indemnity agreements for specific operations without a subsidiary. At present these are negotiated on a case-by-case basis which does not necessarily imply that indemnity will be provided in every situation.

⁸⁷ Archer & Moser-Boehm (2013).

on their ability to operate with negative equity to manage losses should they arise and have been able to continue to operate effectively during extended periods of negative equity.⁸⁸

There are trade-offs associated with these different approaches. Provision by government of sufficient capital to cover a broad range of contingencies to reduce the likelihood a central bank will have to rely on negative equity, and risk adverse effects on credibility, provides operational independence up to the limit of the capital. Tolerance for negative equity arising from policy actions required to achieve the monetary policy mandate is nevertheless an important component. Indemnity agreements may help protect operational credibility. These make explicit the costs to the fiscal authority which enhances transparency. Reliance on negotiating an indemnity agreement during a monetary crisis risks limiting the independence of a central bank's choice and scope to use policy instruments. Therefore, where indemnities are considered necessary, it is probably preferable they are pre-negotiated and have broad coverage across instruments so as not to constrain a central bank's choice of instrument.

Because of the fiscal risks, governments may want to impose a degree of constraint on the extent of funding to cover the potential financial costs associated with quantitative easing. This could be in the form of limiting the amount of capital allocated or that can be accumulated, the tolerance for negative equity, or via the way indemnity agreements are specified. Nevertheless, there is an uneasy trade-off involved protecting the government's fiscal position on the one hand, and potentially impinging on the operational independence of central banks on the other hand. This trade-off arises from constraints on the use of conventional interest rate policy in a low interest environment and the effective lower bound of policy rates.

Quantitative easing as a monetary policy instrument poses a range of reputational risks for central banks and should ideally only be undertaken in the event traditional interest rate instruments are compromised, and if monetary policy is the appropriate policy instrument. As with decisions over policy interest rates, the use of alternative monetary policy instruments should be accompanied by strong accountability requirements to ensure robust decision-making. These requirements include publication of principles to guide the use of these instruments.⁸⁹ A careful assessment of the expected benefits as well as the fiscal costs is also important because while there are potential fiscal costs to quantitative easing, there are also potential fiscal benefits. These benefits include lower costs of public debt and, to the extent the measures support aggregate economic activity, they could increase government net taxes. Although fiscal costs and benefits have offsetting effects in consolidated government balance sheets, only the fiscal costs of interest rate losses land on central bank balance sheets.

Quantitative easing is used to reduce interest rates, and therefore it might be argued there is a 'coincidence of desires' between governing politicians and central bankers, and little need to be concerned with independence. However, there can still be conflicts of interest over the

⁸⁸ See Archer & Moser-Boehm (2013), Bell, *et al.* (2023). The RBA for example, does not rely on indemnities. Recent RBA losses (including unrealised mark-to-market valuation losses) have been partially funded by drawing on reserve funds accumulated from previous retained profits and by moving into negative equity. The RBA is able to continue to operate via its ability to create base money, and has indicated to Government it expects future profits will be retained until its capital is restored (Bullock, 2022).

⁸⁹ An example is Kengmana (2021).

extent of interest rate reductions and influences on yield curves. These would be more acute when governments are increasing spending and borrowing. There may be conflicts over the duration of monetary stimulus, and particularly the timing vis-a-vis general elections. There could be conflicts of interest over the form of monetary stimulus. Quantitative easing involving the purchase of private assets (e.g., equities) have direct resource allocation implications, can involve a potential subsidy if purchased at prices above market value, and have direct wealth distributional effects. This type of credit easing would involve central bankers making inherently political trade-offs and are more appropriately fiscal decisions.

5.7 Fiscal and monetary policy coordination and risks of fiscal dominance

The conventional wisdom that emerged as CBI spread globally and inflation declined, was that macroeconomic stabilisation and inflation targeting should be the responsibility of monetary policy while fiscal policy was considered more appropriate for structural economic priorities.⁹⁰ It was recognised there could be circumstances where fiscal policy might be more effective, such as when monetary policy was compromised by a “liquidity trap” situation or when interest rates are at their effective lower bound.⁹¹ Nevertheless, the early decades of central bank independence were characterised by “monetary dominance” of stabilisation policy.

The features of the pandemic, notably the health impact on labour supply, meant fiscal policy was more efficient at supporting employment and the incomes of vulnerable households and businesses.⁹² Monetary policy had a role to play maintaining financial stability and supporting aggregate demand. However, there was a sense of having to respond quickly and have “all hands to the pump”.⁹³ The pandemic warranted a degree of policy coordination for which central banks and fiscal authorities were not adequately prepared. This raises the question whether there should be more coordination of fiscal and monetary policy in some circumstances.⁹⁴ What are those circumstances? What are the risks to central bank operational independence? How can those risks be mitigated? A central bank cannot dictate what the fiscal response to events should be, although it can explain its policy options and what its response would be in various circumstances, consistent with its legal mandate.

⁹⁰ There were several arguments against the active use of fiscal policy for macroeconomic stabilisation, including potential offsetting private behaviour, implementation lags generating pro-cyclical effects and accentuating cycles, incentives of politicians and time-inconsistent discretionary fiscal policy. These are perennial issues (Alesina, 2000; Barker, *et al.*, 2008; Brook, 2013; De Haan & Gootjes, 2023).

⁹¹ Krugman (2005), De Long & Tyson (2013).

⁹² In the United States for example, the federal government provided large and highly concentrated support in the form of “stimulus checks” sent directly to households (Brunnermeier, 2023). The fiscal support programme implemented in Australia is discussed in detail in Boland (2023).

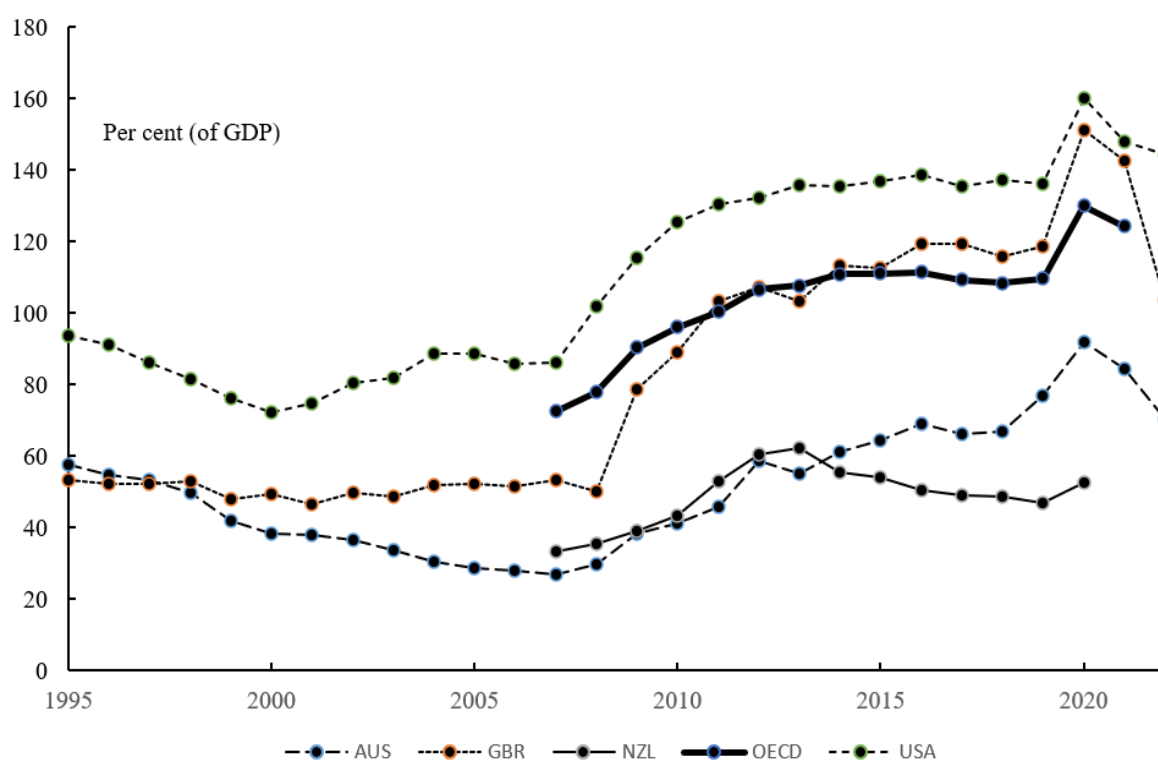
⁹³ The fact there was a combination of fiscal and monetary responses provoked suggestions that a form of “fiscal dominance” was at play. For example, Lingle (2021) described central bank responses to the GFC and the Covid-19 pandemic as indicating they were becoming a “handmaiden to profligate fiscal policy.” In contrast, Haldane’s (2020, p17) interpretation of the circumstances in the UK is that the combination of fiscal and monetary responses did not imply “one policy driving the other, much less of monetary financing of government deficits”.

⁹⁴ This point has also been made by the review of the Reserve Bank of Australia (Australian Government, 2023) which suggests there should be increased joint work between Treasury and RBA on the relative roles of fiscal and monetary policy.

These questions need resolving. How they are resolved could have important implications for central bank operational autonomy. While the empirical evidence reviewed in Section 4.4 indicates central bank independence has, in general, lead to smaller fiscal deficits, the exception tends to hold during election years. Despite central bank independence, fiscal policy has tended to be more expansionary during election years. This suggests the intensity of political interests could vary over the political cycle and impede efficient coordination. This was a factor that motivated transferring responsibility for monetary policy to independent central banks.

The incentives for governments to put pressure on central banks to moderate interest rates and reduce the cost of servicing public debt have risen in recent years. Government debt ratios have been rising over the last two decades. Figure 5 illustrates gross general government debt as a per cent of nominal GDP per year for the total OECD and for a selection of OECD countries, from 1995 to 2022. As mentioned in Section 3, the public debt environment during the first two decades of central bank independence was conducive to enabling “monetary dominance”. Debt levels were not so pressing to induce political pressure on central banks to lower rates and support lower service costs of public debt. This changed during the GFC and the years that followed, and during the pandemic era. There are countries with public debt ratio trends that are exceptions to that for the whole of the OECD but overall, incentives for politicians to pressure central banks to adopt more accommodating interest rate policies have increased.

Figure 5: Gross general government debt ratios: OECD countries, 1995 to 2022



Notes: Data from OECD (2023) General government debt (indicator) doi: 10.1787/a0528cc2-en (Accessed on 25 May 2023). The data are gross debt of general government as a percentage of GDP. Debt is the sum of the following liability categories (as applicable): currency and deposits; debt securities, loans; insurance, pensions and standardised guarantee schemes, and other accounts payable.

Moreover, despite some debt reduction in 2022, the prospect is for debt ratios to rise further in the near term. The OECD (2023) expects debt levels of developed economies to increase slightly in 2023 as many countries aim to protect households and businesses from rising prices and as debt servicing costs increase. The current geopolitical environment is adding further uncertainty over future government funding needs. The OECD reports that developing economies also have historically high levels of debt and a challenging funding environment. The combination of high inflation, high nominal interest rates and high public debt levels was a lethal combination for monetary policy to contend with during the 1970s and 1980s. Similar circumstances could arise if central banks struggle to contain the post-pandemic inflation or central bank independence comes under political pressure.

There are important reasons to consider improvements to the assignment of fiscal and monetary policies in response to destabilising events. However, higher public debt levels increase the risk of “fiscal dominance” and higher inflation highlighted by Sargent and Wallace (1984) and reiterated by Leeper (2023). This risk can come through pressures on central banks to reduce interest rates and public debt servicing costs or via pressure to monetise debt.

6. Conclusion

Central bank independence spread globally after the 1980s and has become a critical foundation for credible monetary policy and sustained low inflation. Other benefits accruing from CBI, include improved financial stability and lower fiscal deficits outside election years. There are qualifications to these conclusions. The effects of CBI can vary by type of political environment, the economic characteristics of countries, and the presence of other economic policy frameworks such as fiscal frameworks and currency regimes.

Despite these benefits, challenges to central bank independence have emerged. The main characteristics and legislated limits to central bank independence include objectives (mandates) and conditions pertaining to objectives, and accountability and transparency requirements. The most critical aspect of central bank independence is discretion over the choice and operation of monetary policy instruments in pursuit of the legislated objectives. This paper suggests there are intersections between legislated objectives and operational independence that could have implications for central bank independence and the political acceptability of CBI. These intersections involve expanding central bank mandates, conditions attached to the pursuit of primary objectives of monetary policy, and funding arrangements that could influence central bank independence over the choice and scope of alternative monetary policy instruments.

Although there have been overt political threats to central bank independence, perhaps the more important challenge is how central banks respond to the criticisms that emerged from the extraordinarily difficult circumstances created by the Covid-19 pandemic, the Russian invasion of Ukraine, and the surge of inflation that followed. There are important lessons from this experience including the choice and application of policy instruments, the scope for improving fiscal and monetary policy coordination while protecting central bank operational independence, and insights into the dynamics of inflation. How central banks respond to this

experience and the insights will influence the acceptability of central bank independence and the determination of monetary policy.

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